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Question

In a three-year study of parasitic infections by *Anomotaenia brevis* tapeworms in *Temnothorax nylanderi* ants, entomologist Susanne Foitzik and colleagues found something unexpected: rather than reducing its host's fitness, as is typical of parasites, *A. brevis* greatly extends the lifespan of a *T. nylanderi* worker ant and seems to halt the effects of aging. Furthermore, those infected receive special treatment, ceasing their share of labor to sustain the colony and remaining in the nest as uninfected workers feed, groom, and transport them. By contrast, the researchers observed that uninfected workers in parasitized colonies have shortened lifespans, most likely because the _____

Which choice most logically completes the text?

Answer

- A. uninfected workers are at high risk for direct exposure to *A. brevis* in the course of providing social care to the infected workers in the nest.
- B. need to compensate for reduced contributions within the colony while also caring for infected workers is burdensome to the uninfected workers.
- C. high level of activity maintained by the uninfected workers makes them better able than infected workers to quickly disperse when the nest is attacked by a predator.
- D. average lifespan of *T. nylanderi* worker ants in colonies without parasitic activity typically falls well below three years, the range covered by the study.